D.5.9. std::once_flag class

Instances of `std::once_flag` are used with `std::call_once` to ensure that a particular function is called exactly once, even if multiple threads invoke the call concurrently.

Instances of `std::once_flag` are not `CopyConstructible, CopyAssignable, Move-Constructible`, or `MoveAssignable`.

Class definition

```
struct once_flag
{
    constexpr once_flag() noexcept;

    once_flag(once_flag const& ) = delete;
    once_flag& operator=(once_flag const& ) = delete;
};
```

Std::Once_Flag Default Constructor

The `std::once_flag` default constructor creates a new `std::once_flag` instance in a state, which indicates that the associated function hasn't been called.

Declaration

```
constexpr once_flag() noexcept;
```

Effects

Constructs a new `std::once_flag` instance in a state, which indicates that the associated function hasn't been called. Because this is a `constexpr` constructor, an instance with static storage duration is constructed as part of the static initialization phase, which avoids race conditions and order-of-initialization problems.

D.5.10. std::call_once function template